WONG HOK KAN

黃學勤

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PROFILE

Third-year Electronic Engineering student at the Hong Kong University of Science and Technology with hands-on experience in computer networking, systems programming, and circuit work. Currently building a user-space TCP implementation in C++ and contributing to open-source projects including Activist, Kinto (Mozilla Services), and OffPDF. Open to internships in software engineering, computer networking, and hardware, including circuit design.

EDUCATION

The Hong Kong University of Science and Technology Hong Kong

BEng in Electronic Engineering Sep 2024 - May 2028 (expected)

- Relevant coursework: Computer Networks (ELEC 3120).

OPEN SOURCE

- Oct 2026 - Activist.org: upgraded Vue to 3.5.42 to remediate a high-severity XSS advisory and aligned the Vitest coverage provider; verified with 1,906 unit tests, lint, and type checks (PRs #2457 and #2464, merged).
- Oct 2026 - Kinto (Mozilla Services): repaired docs builds and documented PostgreSQL index tuning; verified with tests and local docs builds (PRs #3803, #3802, merged).
- Oct 2026 - OffPDF (Rust/Tauri): authored the local-AI licensing matrix used in their runtime and model research, verifying licenses and primary sources for 11 candidates (PR #127, merged).

TECHNICAL PROJECTS

FoggyTCP: A User-Space TCP Implementation Sep 2026 - Present

ELEC 3120 Computer Networks, course project · C++, Linux, Git, Vagrant, VirtualBox

- Developing a user-space TCP implementation in C++, covering sliding-window transmission, loss recovery, and congestion control.
- Configured an isolated Linux development environment with Vagrant and VirtualBox for building and testing the protocol.
- Using Git for version control and Linux development tools for building, testing, and debugging the implementation.

SKILLS

Programming: C, C++, Python

Networking: TCP/IP fundamentals, socket programming

Development tools: Linux, Git, GitHub pull requests, Vagrant, VirtualBox

AI-assisted development: Codex for prototyping and debugging

Hardware and laboratory: Arduino, circuit design, oscilloscope, MATLAB

Languages: Cantonese, English, Mandarin